

the horizontal lines must be larger than the  $d$  for the vertical lines.

19 C

Option A :  $F = \frac{q^2}{4\pi\epsilon_0 r^2}$ .  $r$  decreases and hence, force increase. Option A is incorrect.

Option B : Electric field strength is a vector quantity. Sketching the individual  $E$  due to each charge at O and summing vectorially to find the resultant shows the  $E$  has decreased. Option B is incorrect.

Option D: Electric potential is a scalar quantity and hence resultant potential at O remains zero even when the change is made. Hence option D is incorrect.

Option C :  $U = -\frac{q^2}{4\pi\epsilon_0 r}$ .  $r$  decrease,  $U$  becomes more negative. Hence  $U$  decreases. Option C is correct.

(Alternatively you can think of the fact that the charges being closer now experience a stronger attractive force and hence will be at lower potential energy and will require a greater amount of energy to separate them to infinity.)

20 A

$dV$